

INDEPENDENT EVENTS:

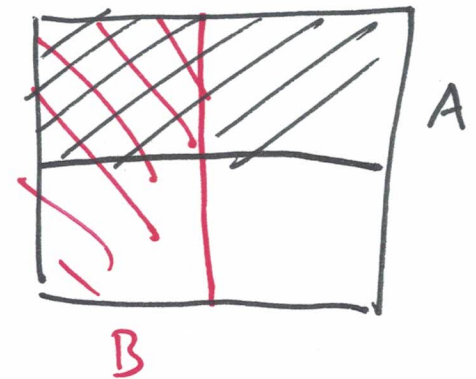
TWO EVENTS ARE INDEPENDENT IF

$$P(A \cap B) = P(A) \cdot P(B)$$

RECALL $P(A|B) = \frac{P(A \cap B)}{P(B)}$

SO FOR INDEPENDENT EVENTS

$$P(A|B) = P(A)$$



DRAW ONE CARD FROM NORMAL DECK:

A: CARD IS RED

B: CARD IS A KING OR A HEART.

$$P(A) = \frac{1}{2} = \frac{26}{52}$$

$$P(B) = \frac{13 + 3}{52} = \frac{16}{52} = \frac{4}{13}$$

$$P(A \cap B) = P(\text{CARD IS RED AND A KING OR A HEART})$$

$$\frac{13 + \cancel{1}}{52} = \frac{14}{52} = \frac{7}{26} \stackrel{?}{=} \frac{1}{2} \cdot \frac{4}{13}$$

$\Rightarrow$ NOT INDEPENDENT.

$$\neq P(A) \cdot P(B)$$

DRAW 2 CARDS, NO REPLACEMENT

A: FIRST CARD IS A KING = $\frac{1}{13}$

B: SECOND CARD IS A HEART

$$\begin{array}{ccccccc} \rightarrow & \frac{1}{4} & \cdot & \frac{12}{51} & + & \frac{3}{4} & \cdot & \frac{13}{51} & = & \frac{12+39}{4 \cdot 51} & = & \frac{1}{4} \\ & \heartsuit & & \heartsuit & & \times & & \heartsuit & & & & \end{array}$$

$P(A \cap B)$

$$\frac{1}{52} \cdot \frac{12}{51} + \frac{3}{52} \cdot \frac{13}{51} = \frac{12+39}{52 \times 51} = \frac{1}{52}$$

K $\heartsuit$

K $\times$

$$= \frac{1}{13} \cdot \frac{1}{4}$$

ARE INDEPENDENT!

$$= P(A) \cdot P(B)$$

BERNOULLI TRIALS:

SEQUENCE OF I.I.D. BINARY EXPERIMENTS.

I.I.D. = INDEPENDENTLY AND IDENTICALLY
DISTRIBUTED.

BERNOULLI'S THEOREM:

$$P\left(\left|\frac{k}{n} - p\right| < \varepsilon\right) > 1 - \frac{p(1-p)}{n\varepsilon^2}$$

SUCCESSSES

TOLERANCE

TRIALS

UNDERLYING
PROB OF SUCCESS



$$P(\text{CATCH FISH}) = \frac{2}{5}$$

7500 TRIES

$$P(\text{FISH} \in [2900, 3100])$$

$$\text{WE EXPECT } 7500 \cdot \frac{2}{5} = \boxed{3000}$$

$$P\left(\left|k - \overset{3000}{np}\right| < \overset{100}{n\varepsilon}\right)$$

↑
ACTUAL
OF
SUCCESSSES

↑
EXPECTED
SUCCESSSES

$$n\varepsilon = 100$$

$$\varepsilon = \frac{100}{7500} = \frac{1}{75}$$

$$> 1 - \frac{\frac{2}{5} \cdot \frac{3}{5}}{7500 \cdot \left(\frac{1}{75}\right)^2}$$

$$= 0.82$$

